

SOME FURTHER PROPERTIES OF TEST CRITERIA IN MULTIVARIATE ANALYSIS*

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1. Introduction and Summary

Consider two sets of variables $X^{(1)}: q \times 1$ and $X^{(2)}: r \times 1, q+r = p$. We are interested in linear relationships between these two sets of variables and we will distinguish between the following two cases:

Case 1:

Here $X^{(2)}$ is a set of predetermined (or fixed) variables and the following relationships exists between $X^{(1)}$ and $X^{(2)}$.

$$(1.1) \quad E(X^{(1)}) = \beta X^{(2)}$$

or

$$X^{(1)} = \beta X^{(2)} + U$$

where $E(U) = 0$.

Here $\beta: q \times r$ is an unknown matrix to be estimated and $U: q \times 1$ is the error vector.

In this paper we make the assumption that the errors are normally distributed, that is U is distributed multivariate normal $N(0, \Phi)$ or, equivalently, $X^{(1)}$ is distributed multivariate normal $N(\beta X^{(2)}, \Phi)$.

Case 2:

Here the vector X where $X' = (X^{(1)'}, X^{(2)'})$ has a p -variate normal distribution $N(0, \Sigma)$ and the conditional distribution of $X^{(1)}$ given $X^{(2)}$ is $N(\beta X^{(2)}, \Sigma_{11.2})$.

The conditional mean is the regression function

$$(1.3) \quad E(X^{(1)}/X^{(2)}) = \beta X^{(2)}$$

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and the conditional variance-covariance matrix is $\Sigma_{11.2}$. If

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \text{ with } \Sigma_{11}:q \times q, \Sigma_{12}:q \times r$$

and $\Sigma_{22}:r \times r$ then from Anderson (1958) follows that

$$(1.4) \quad \beta = \Sigma_{12} \Sigma_{22}^{-1}$$

and

$$(1.5) \quad \Sigma_{11.2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

Case 1 is clearly a special case of Case 2. The purpose of this paper is to study these two models in detail and to discuss tests of significance on the β 's. The test criteria employed will be considered in section (2) and extensions to three sets will also be considered in section (2). The distributions of some of these statistics will be considered in section (3).

2. Test criteria in Multivariate Analysis

Several measures have been defined in the literature to measure the relationship between the two sets $X^{(1)}$ and $X^{(2)}$. All these measures depend on the generalised multiple correlation matrix (Troskie (1969), (1971a)) which will be defined below.

Suppose that in case 1 fixed vectors $X_{(1)}^{(2)}, \dots, X_{(n)}^{(2)}$ are taken and the corresponding vector observations $X_{(1)}^{(1)}, \dots, X_{(n)}^{(1)}$ are observed. In case 2 a sample $X_{(\alpha)}, \alpha = 1, \dots, n$ is taken from $N(0, \Sigma)$. Let

$$(2.1) \quad A_{ij} = \sum_{\alpha=1}^n X_{(\alpha)}^{(i)} X_{(\alpha)}^{(j)'}; i, j = 1, 2$$

and let

$$(2.2) \quad A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \text{ with } A_{11}:q \times q, A_{12}:q \times r \text{ and } A_{22}:r \times r, q \leq r.$$

The generalized multiple correlation matrix R was defined by Khatri (1964) as

$$(2.3) \quad R = A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}$$

where $A_{11}^{1/2}$ is the positive definite square root of A_{11} and can either be symmetric or lower-triangular. We also adopt the convention that in an expression like $A^{1/2} B A^{1/2}$ the post-multiplier is $(A^{1/2})'$.

The matrix R has the desirable property that R can be interpreted as a measure of the correlation between $X^{(1)}$ and $X^{(2)}$ for case 2, and in case 1, as a measure of the linear influence of $X^{(2)}$ and $X^{(1)}$. Similarly $1-R = A_{11}^{-1/2} A_{11.2} A_{11}^{-1/2}$ where $A_{11.2} = A_{11} - A_{12} A_{22}^{-1} A_{21}$ can be interpreted as a measure of the unexplained variations in the regression of $X^{(1)}$ on $X^{(2)}$. The characteristic roots $0 < r_1^2 < \dots < r_q^2 < 1$ of R for case 2, are also the squares of the canonical correlation coefficients.

In case 1 one would like to test the hypothesis

$$(2.4) \quad H_0 : \beta = 0$$

that is whether $X^{(1)}$ can be estimated from $X^{(2)}$. In case 2 where $\beta = \Sigma_{12} \Sigma_{22}^{-1}$, testing (2.4) is equivalent to testing whether $X^{(1)}$ and $X^{(2)}$ are independent. The test criteria proposed are $W = |1-R|$ (Wilks likelihood ratio, Wilks (1932), Anderson (1958)), Roy's largest and smallest root criterion, i.e. r_q^2 and r_1^2 (Roy (1957)), Hotelling's T_0^2 or Pillai's $U^{(q)} = T_0^2/c = \text{tr } R(1-R)^{-1}$ (Pillai 1955), Pillai's V and Q where

$$V = \text{tr } R \text{ and } Q = \text{tr } R^{-1} (1-R) \text{ (Pillai (1955))},$$

and the trace correlation coefficient

$$\bar{r}^2 = \frac{1}{q} \text{tr } R = \frac{1}{q} V \text{ (Hooper (1959), Troskie (1971b)).}$$

Several authors have already investigated the distributional properties of the statistics W , r_q^2 (largest root), r_1^2 (smallest root), U , V and Q . The central distribution of W has been investigated by Anderson (1958); the noncentral distribution of W for both case 1 and case 2 by Pillai, Al-Ani and Jouris (1969).

The central and noncentral distributions of the largest root criterion r_q^2 have been investigated by Pillai (1955, 1956a, 1956b, 1957, 1960, 1964, 1965, 1967), Pillai and Bantegui (1959), Foster and Rees (1957), Foster (1957, 1958) and Sugiyama (1967).

Khatri and Pillai (1968) derived an explicit expression for the distribution of r_q^2 for case 1 while De Waal (1969) derived the noncentral distribution of r_q^2 for case 2.

The central distribution of the smallest root r_1^2 has been investigated by Pillai (1956) and Sugiyama (1967). The noncentral distributions and moments of the smallest root (1965a) for both case 1 and case 2 will be given in Section 3 of the present paper.

The central and noncentral distributions of U and V for case 1 have been investigated by Pillai (1955, 1957, 1960), Pillai and Samson (1959), Pillai and Jayachandran (1967), Constantine (1966), Khatri and Pillai (1968) and De Waal (1968). The noncentral distributions of U and V for case 2 have been derived by Troskie (1971a).

The central and noncentral distributions of the statistic Q will also be given in section 3.

Except for the statistics Q and r_q power comparisons of the above statistics have been made by Pillai and Jayachandran (1967, 1968) for the case where $q = 2$ (see also Troskie (1971a, 1971b)).

Extensions to three sets of variables ($X^{(1)}, X^{(2)}, X^{(3)}$) with q, r and s components respectively, are immediate. Measures of the relationships that exist between two sets, say $X^{(1)}$ and $X^{(2)}$ when the third set, say $X^{(3)}$ is fixed at some value $x^{(3)}$ are of considerable theoretical and practical interest. Tests of conditional linear relationships can be constructed.

As was shown by Juritz (1971) all the above test criteria are relevant and all these criteria are based on the partial generalised correlation matrix, which has the same distributional properties as that of the generalised multiple correlation matrix R with the sample size N replaced by $N-s$ (where s is the number of variables held fixed). Hence all *partial test criteria* have the same distributions as the above test criteria except that N is replaced by $N-s$. Of particular importance is the smallest partial canonical correlation coefficient which will have the same distribution as the smallest canonical correlation coefficient with N replaced by $N-s$.

3. The distribution and moments of the smallest root r_1^2 and $Q = \text{tr } R^{-1}(I-R)$

The distributions of r_1^2 and $Q = \text{tr } R^{-1}(I-R) = \sum_{i=1}^q (1-r_i^2)$ can be derived from the joint distribution of the roots $r_1^2, \dots, r_q^2$, $0 < r_1^2 < r_2^2 < \dots < r_q^2 < 1$ of R . The joint density of $r_1^2, \dots, r_q^2$ is given by (Constantine (1963));

Case 1

$$(3.1) \quad (\Pi^{1/2} q^2 / \Gamma_q(1/2q)) \beta(R, 1/2r, 1/2n-1/2r) e^{-1/2 \text{tr} \Omega} \alpha_q(R) {}_1F_1(1/2n; 1/2r; 1/2\Omega, R), \quad 0 \leq r_1^2 \leq r_2^2 \leq \dots \leq r_q^2 \leq 1$$

where

$$(3.2) \quad \beta(R, \frac{1}{2}r, \frac{1}{2}n - \frac{1}{2}r) = (\Gamma_q(\frac{1}{2}n) / \Gamma_q(\frac{1}{2}r) \Gamma_q(\frac{1}{2}n - \frac{1}{2}r)) |R|^{\frac{1}{2}(r-q-1)} \\ |I-R|^{\frac{1}{2}(n-r-q-1)}, 0 < R < I,$$

$$(3.3) \quad \alpha_q(R) = \prod_{i > j} (r_i^2 - r_j^2),$$

$$(3.4) \quad \Omega = \Phi^{-1} \beta A_2 \beta'$$

$$(3.5) \quad \Gamma_q(\frac{1}{2}n) = \prod_{i=1}^q (q-i) \Gamma(\frac{1}{2}(n+1-i))$$

and the hypergeometric function has been defined by James (1964);

Case 2

$$(3.6) \quad (\prod_{i=1}^q q^2 / \Gamma_q(\frac{1}{2}q)) \beta(R, \frac{1}{2}r, \frac{1}{2}n - \frac{1}{2}r) |I-P|^{\frac{1}{2}n} \alpha_q(R) {}_2F_1(\frac{1}{2}n, \frac{1}{2}n; \frac{1}{2}r; P, R), \\ 0 \leq r_1^2 \leq r_2^2 \leq \dots \leq r_q^2 \leq 1$$

where $P = \text{diag}(\rho_1^2, \dots, \rho_q^2)$ and $\rho_1^2, \dots, \rho_q^2$ are the squares of the population canonical correlation coefficients.

The following results are of importance.

Lemma 3.1 (Khatrı and Pillai (1968))

If A_k denotes the integral

$$(3.7) \quad \int_D |Z|^{\frac{1}{2}(f-q-1)} \alpha_q(Z) C_k(Z) dz_1, \dots, dz_{q-1}$$

where

$$Z = \text{diag}(z_1, \dots, z_q), z_q = I - z_1 - \dots - z_{q-1}$$

and D is given by

$$(3.8) \quad D: (0 < z_1 < z_2 < \dots < z_{q-2} < z_{q-1} < z_q = I - z_1 - z_2 - \dots - z_{q-1}),$$

then

$$(3.9) \quad A_{\kappa} = ((\frac{1}{2}f)_{\kappa} \Gamma_q(\frac{1}{2}q) \Gamma_q(\frac{1}{2}f) C_{\kappa}(I_q) / \Pi^{\frac{1}{2}q^2} (\frac{1}{2}fq)_{\kappa} \Gamma(\frac{1}{2}fq)).$$

The zonal polynomial $C_{\kappa}(Z)$ and the hypergeometric coefficient $(a)_{\kappa}$ are defined in Constantine (1963).

Lemma 3.2 (Troskie (1971a))

For symmetric matrix S

$$(3.10) \quad |I+S|^{-a} (a)_{\kappa} C_{\kappa}((I+S)^{-1}) \\ = \sum_{m=0}^{\infty} \sum_{\eta} \sum_{\delta} ((-1)^m (a)_{\delta} g_{\kappa, \eta}^{\delta} C_{\eta}(S) C_{\delta}(I) / C_{\eta}(I) m!) ,$$

where $\kappa = (k_1, \dots, k_q)$, $k_1 \geq k_2 \geq \dots \geq k_q \geq 0$, $\sum k_i = k$ is a partition of the integer k into not more than q parts, $\eta = (m_1, \dots, m_q)$, $m_1 \geq m_2 \geq \dots \geq m_q \geq 0$, $\sum m_i = m$ is a partition of the integer m into not more than q parts, $g_{\kappa, \eta}^{\delta}$ is the coefficient of $C_{\delta}(Z)$ in the product $C_{\kappa}(Z) C_{\eta}(Z)$ and $\delta = (\delta_1, \dots, \delta_q)$, $\delta_1 \geq \dots \geq \delta_q \geq 0$, $\sum \delta_i = m+k = d$. The coefficient $g_{\kappa, \eta}^{\delta}$ has been tabulated by Khatri and Pillai (1968) for various values of the arguments.

Lemma 3.3 (Sugiyama (1967))

Let Λ_{ℓ} be a diagonal matrix with diagonal elements $1 > \lambda_2 > \dots > \lambda_q > 0$ and let κ be a partition of k . Then

$$(3.11) \quad \int_{1 > \lambda_2 > \dots > \lambda_q} |\Lambda_{\ell}|^{t-\frac{1}{2}(q+1)} C_{\kappa}(I \Lambda_{\ell}) |I_{q-1} \Lambda_{\ell}|^{\alpha'_{q-1}} (\Lambda_{\ell}) d\lambda_2, \dots, d\lambda_q \\ = ((q+1+k) \Gamma_q(q/2) \Gamma_q(1) (t)_{\kappa} \Gamma_q(\frac{1}{2}(q+1)) C_{\kappa}(I) / \Pi^{\frac{1}{2}q^2} \Gamma_q(t+\frac{1}{2}(q+1)) (t+\frac{1}{2}(q+1))_{\kappa})$$

where

$$3.12 \quad I \Lambda_{\ell} = \begin{vmatrix} 1 & 0 \\ 0 & \Lambda_{\ell} \end{vmatrix} \text{ and } \alpha'_{q-1} = \prod_{i < j}^{q-1} (\lambda_i - \lambda_j).$$

Theorem 3.1

The density of the smallest root r_1^2 of R is given by:

Case 1

$$(3.13) \quad f_1(r_1^2) = (\Gamma_q(\frac{1}{2}n) \Gamma_q(\frac{1}{2}(q+1)) / \Gamma_q(\frac{1}{2}r) \Gamma_q(\frac{1}{2}(n-r+q+1))) \\ e^{-tr \frac{1}{2} \Omega} \sum_{\kappa} \sum_{\kappa} \sum_t \sum_r \sum_{\eta \leq \kappa} \sum_{\delta} ((\frac{1}{2}n)_{\kappa} \alpha_{\tau \rho \tau, \eta}^{\delta} (1-r_1^2)^{\frac{1}{2}q(n-r)+d-1} \\ \cdot (\frac{1}{2}q(n-r)+d) (\frac{1}{2}(q+1-r))_{\tau} (\frac{1}{2}(n-r))_{\delta} C_{\kappa}(\frac{1}{2}\Omega) C_{\delta}(I) / \\ \cdot (\frac{1}{2}r)_{\kappa} (\frac{1}{2}(n-r+q+1))_{\kappa} C_{\kappa}(I) k! t!), 0 < r_1^2 < 1.$$

Case 2

$$3.14 \quad f_2(r_1^2) = (\Gamma_q(\frac{1}{2}n) \Gamma_q(\frac{1}{2}(q+1)) / \Gamma_q(\frac{1}{2}r) \Gamma_q(\frac{1}{2}(n-r+q+1)))$$

$$|I-P|^{\frac{1}{2}n} \sum_k \sum_{\kappa} \sum_t \sum_{\tau} \sum_{\eta \leq \kappa} \sum_{\delta} (\frac{1}{2}n)_{\kappa} (\frac{1}{2}n)_{\kappa} \alpha_{\eta}^{\delta} e_{\tau}^{\delta}$$

$$(1-r_1^2)^{\frac{1}{2}q(n-r)+d-1} (\frac{1}{2}q(n-r)+d) (\frac{1}{2}(q+1-r))_{\tau} (\frac{1}{2}(n-r))_{\delta}$$

$$C_{\kappa}(P) C_{\delta}(I) / (\frac{1}{2}r)_{\kappa} (\frac{1}{2}(n-r+q+1))_{\delta} C_{\kappa}(I) k! \quad t! \quad), \quad 0 < r_1^2 < 1$$

Proof

The derivations of (3.13) and (3.14) are similar and thus only the derivation of (3.14) will be given. The joint density of the roots $r_1^2, \dots, r_q^2$, $0 < r_1^2 < \dots < r_q^2 < 1$ of R for case 2, is given by (3.6). Let $L = I - R$, and let the roots of L be $1 > \ell_1 > \ell_2 > \dots > \ell_q > 0$. The joint density of $\ell_1, \dots, \ell_q$ is then given by

$$(3.15) \quad f(\ell_1, \dots, \ell_q) = (\pi^{q^2/2} \Gamma_q(\frac{1}{2}q)) \beta(L, \frac{1}{2}n - \frac{1}{2}r, \frac{1}{2}r) |I-P|^{\frac{1}{2}n} \alpha'_q(L) \cdot {}_2F_1(\frac{1}{2}n, \frac{1}{2}n; \frac{1}{2}r; P, I-L)$$

where $\alpha'_q(L) = \prod_{i < j}^q (\ell_i - \ell_j)$, $0 \leq \ell_q \leq \dots \leq \ell_1 \leq 1$.

Since $\ell_1 = 1 - r_1^2, \dots, \ell_q = 1 - r_q^2$, we will first derive the density of ℓ_1 , which is the largest root of L . Let $\lambda_i = \ell_i / \ell_1$, $i = 2, \dots, q$ and let $\Lambda_q = \text{diag}(\lambda_2, \dots, \lambda_q)$, ${}^1\Lambda_q = \begin{vmatrix} 1 & 0 \\ 0 & \Lambda_q \end{vmatrix}$.

Then $|L| = e^q |\Lambda_q|$ and the Jacobian is ℓ_1^{q-1} .

Using

$$(3.16) \quad |I-L|^{\frac{1}{2}(r-q-1)} C_{\kappa}(I-L) = \sum_{t=0}^{\infty} \sum_{\tau} (\frac{1}{2}(q+1-r))_{\tau} (C_{\tau}(L)/t!) \sum_{\eta \leq \kappa} \alpha_{\eta}^{\delta} C_{\eta}(L) \\ = \sum_{t=0}^{\infty} \sum_{\tau} \sum_{\eta \leq \kappa} \sum_{\delta} (\alpha_{\eta}^{\delta} e_{\tau}^{\delta} / \eta C_{\delta}(L)/t!)$$

where

$$C_{\tau}(L) C_{\eta}(L) = \sum_{\delta} e_{\tau, \eta}^{\delta} C_{\delta}(L), \quad C_{\kappa}(I-L) = \sum_{\eta \leq \kappa} \alpha_{\eta} C_{\eta}(L)$$

and δ is a partition of the integer $d = t+k$ into not more than q parts, we can write the joint density of ℓ_1 and Λ_q as

$$(3.17) \quad f(\ell_1, \Lambda_\ell) = (\Pi^{q^2/2} / \Gamma_q(\frac{1}{2}n) / \Gamma_q(\frac{1}{2}n) / \Gamma_q(\frac{1}{2}n - \frac{1}{2}r) \Gamma_q(\frac{1}{2}r))$$

$$|I-P|^{\frac{1}{2}n} \ell_1^{\frac{1}{2}(n-r)q-1} |\Lambda_\ell|^{\frac{1}{2}(n-r-q-1)} |I-\Lambda_\ell|^{\alpha'_{q-1}} (\Lambda_\ell)$$

$$\cdot \Sigma_k \Sigma_\kappa \Sigma_{\eta \leq \kappa} \Sigma_t \Sigma_\tau \Sigma_\delta ((\frac{1}{2}n)_\kappa \ell_1^d \alpha_{\eta \tau}^\delta \eta C_\delta (1 \Lambda_\ell) C_\kappa(P) (\frac{1}{2}(q+1-r))_\tau /$$

$$(\frac{1}{2}r)_\kappa C_\kappa(I) t! \quad k! \quad)$$

Integrating (3.17) over Λ_ℓ using Lemma 3.3 yields the density of ℓ_1 as

$$(3.18) \quad (\Gamma_q(\frac{1}{2}n) \Gamma_q(\frac{1}{2}(q+1)) / \Gamma_q(\frac{1}{2}r) \Gamma_q(\frac{1}{2}(n-r+q+1))) |I-P|^{\frac{1}{2}n} \Sigma_k \Sigma_\kappa \Sigma_t \Sigma_\tau \Sigma_{\eta \leq \kappa} \Sigma_\delta$$

$$((\frac{1}{2}n)_\kappa (\frac{1}{2}n)_\kappa \alpha_{\eta \tau}^\delta \ell_1^{\frac{1}{2}q(n-r)+d-1} (\frac{1}{2}q(n-r)+d) (\frac{1}{2}(q+1-r))_\tau (\frac{1}{2}(n-r))_\delta C_\kappa(P)$$

$$C_\delta(I) / (\frac{1}{2}r)_\kappa (\frac{1}{2}(n-r+q+1))_\delta C_\kappa(I) k! \quad t! \quad), 0 < \ell_1 < 1.$$

Therefore, by substituting $1-r_1^2$ for ℓ_1 we have the density of the smallest root r_1^2 of R , namely the formula (3.14).

Theorem 3.2

The cdf of the smallest root r_1^2 of R is given by

Case 1

$$(3.19) \quad P(r_1^2 < x) = 1 - P(r_1^2 > x)$$

$$= 1 - (\Gamma_q(\frac{1}{2}n) \Gamma_q(\frac{1}{2}(q+1)) / \Gamma_q(\frac{1}{2}r) \Gamma_q(\frac{1}{2}(n-r+q+1)))$$

$$e^{-tr \frac{1}{2} \Omega} \Sigma_k \Sigma_\kappa \Sigma_t \Sigma_\tau \Sigma_{\eta \leq \kappa} \Sigma_\delta ((\frac{1}{2}n)_\kappa \alpha_{\eta \tau}^\delta \eta (1-x)^{\frac{1}{2}q(n-r)+d} (\frac{1}{2}(q+1-r))_\tau (\frac{1}{2}(n-r))_\delta$$

$$C_\kappa(\frac{1}{2} \Omega) C_\delta(I) / (\frac{1}{2}r)_\kappa (\frac{1}{2}(n-r+q+1))_\delta C_\kappa(I) k! \quad t! \quad)$$

Case 2

$$(3.20) \quad P(r_1^2 < x) = 1 - P(r_1^2 > x)$$

$$= 1 - (\Gamma_q(\frac{1}{2}n) \Gamma_q(\frac{1}{2}(q+1)) / \Gamma_q(\frac{1}{2}r) \Gamma_q(\frac{1}{2}(n-r+q+1)))$$

$$|I-P|^{\frac{1}{2}n} \Sigma_k \Sigma_\kappa \Sigma_t \Sigma_\tau \Sigma_{\eta \leq \kappa} \Sigma_\delta ((\frac{1}{2}n)_\kappa (\frac{1}{2}n)_\kappa \alpha_{\eta \tau}^\delta \eta (1-x)^{\frac{1}{2}q(n-r)+d}$$

$$(\frac{1}{2}(q+1-r))_\tau (\frac{1}{2}(n-r))_\delta C_\kappa(P) C_\delta(I) / (\frac{1}{2}r)_\kappa (\frac{1}{2}(n-r+q+1))_\delta C_\kappa(I) k! \quad t! \quad).$$

When $X^{(1)}$ and $X^{(2)}$ are not linearly related in case 1, i.e. $\beta=0$ or independent in case 2, i.e. $\Sigma_{12} = 0$ ($P=0$) then the densities (3.13) and (3.14) reduce to

$$(3.21) \quad (\Gamma_q(\frac{1}{2}n)\Gamma_q(\frac{1}{2}(q+1))/\Gamma_q(\frac{1}{2}r)\Gamma_q(\frac{1}{2}(n-r+q+1)))\Sigma_t\Sigma_\tau^{\frac{1}{2}(q(n-r)+t)(\frac{1}{2}(q+1-r))}_\tau \\ \cdot (\frac{1}{2}(n-r))_\tau C_\tau(I)(1-r_1^2)^{\frac{1}{2}q(n-r)+t-1}/(\frac{1}{2}(n-r+q+1))_\tau t^{\frac{1}{2}q(n-r)+t-1}$$

which was first derived by Sugiyama (1967).

Theorem 3.3

The h th moment of the smallest root r_1^2 is given by

Case 1

$$(3.22) \quad E((1-r_1^2)^h) = (\Gamma_q(\frac{1}{2}n)\Gamma_q(\frac{1}{2}(q+1))/\Gamma_q(\frac{1}{2}r)\Gamma_q(\frac{1}{2}(n-r+q+1))) \\ \cdot e^{-tr^{\frac{1}{2}\Omega}\Sigma_K\Sigma_K\Sigma_t\Sigma_{\eta\leq K}\Sigma_\delta((\frac{1}{2}n)_K\alpha_{\eta\delta\tau,\eta}^\delta(\frac{1}{2}(q+1-r))_\tau(\frac{1}{2}(n-r))_\delta(\frac{1}{2}q(n-r)+d))} \\ C_K(\frac{1}{2}\Omega)C_\delta(I)/(\frac{1}{2}r)_K(\frac{1}{2}(n-r+q+1))_\delta C_K(I)k! t! (\frac{1}{2}(q(n-r)+d+h))$$

Case 2

In the moments of case 1 replace $e^{-tr^{\frac{1}{2}\Omega}}$ with $(I-P)^{\frac{1}{2}n}$, $C_K(\frac{1}{2}\Omega)$ with $C_K(P)$ and add the coefficient $(\frac{1}{2}n)_K$ in the summation.

The density of $Q = \text{tr } R^{-1}(I-R)$ for case 2 has been derived by Troskie (1971a). The derivations of the density for case 1 is similar and it is easy to show that the density is

$$(3.23) \quad (\Gamma_q(\frac{1}{2}n)/\Gamma_q(\frac{1}{2}r)\Gamma_q(\frac{1}{2}(n-r)q))e^{-tr^{\frac{1}{2}\Omega}Q^{\frac{1}{2}(n-r)q-1}}\sum_{k=0}^{\infty}\sum_{m=0}^{\infty}\Sigma_K\Sigma_\eta\Sigma_\delta((-1)^m \\ \cdot (\frac{1}{2}n)_\delta(\frac{1}{2}(n-r))_\eta Q_{\eta\delta}^m C_K(\frac{1}{2}\Omega)C_\delta(I)/(\frac{1}{2}(n-r)q)_m(\frac{1}{2}r)_K C_K(I)m! k!)$$

for $|Q| < 1$.

To derive the moments of Q we need

Lemma 3.4 (Constantine (1963)).

Let Z be a complex symmetric matrix whose real part is positive definite and let T be an arbitrary complex symmetric matrix. Then

$$(3.24) \quad \int_{S>0} e^{-tr ZS} |S|^{a-\frac{1}{2}(q+1)} C_K(ST)dS \\ = \Gamma_q(a/a)_K |Z|^{-a} C_K(TZ^{-1})$$

the integration being over the space of positive definite $q \times q$ matrices and valid for all complex numbers a satisfying $R(a) > \frac{1}{2}(q-1)$.

Lemma 3.5

For symmetric T

$$(3.25) \int_{S>0} e^{-trS} |S|^{a-\frac{1}{2}(q+1)} C_{\kappa}(TS) C_{\mu}(S) dS \\ = (C_{\kappa}(T)/C_{\kappa}(I)) \Sigma_{\delta} g_{\kappa,\mu}^{\delta} \Gamma_q(a/a)_{\delta} C_{\delta}(I)$$

Proof

The integral is a symmetric function in T . Making the transformation $T \rightarrow HTH'$ where H is a random orthogonal matrix with unit measure, i.e. $\int_{0(q)} dH = 1$ the integral equals

$$\int_{B>0} \int_{0(q)} e^{-trS} |S|^{a-\frac{1}{2}(q+1)} C_{\kappa}(HTH'S) C_{\mu}(S) dH dS.$$

Now using (Constantine (1963))

$$\int_{0(q)} C_{\kappa}(HTH'S) dH = C_{\kappa}(T) C_{\kappa}(S)/C_{\kappa}(I), \text{ and}$$

$$C_{\kappa}(S) C_{\mu}(S) = \Sigma_{\delta} g_{\kappa,\mu}^{\delta} C_{\delta}(S)$$

and then integrating over S using (3.24) yields the result.

Here $\mu = (m_1, \dots, m_q)$, $m_1 \geq m_2 \geq \dots \geq m_q \geq 0$, $\Sigma m_i = m$ is a partition of the integer m into not more than q parts,

$$\delta = (\delta_1, \dots, \delta_q), \delta_1 \geq \delta_2 \geq \dots \geq \delta_q \geq 0, \Sigma \delta_i = m+k.$$

Theorem 3.4

The moment generating function of Q is:

Case 1

$$(3.26) M_I(t) = E(e^{tQ}) = (\Gamma_q(\frac{1}{2}n)/\Gamma_q(\frac{1}{2}r)) (-I)^{-\frac{1}{2}(n-r)} q t^{-\frac{1}{2}(n-r)} q \\ e^{-tr\frac{1}{2}\Omega} \Sigma_m \Sigma_k \Sigma_{\mu} \Sigma_{\kappa} \Sigma_{\delta} ((\frac{1}{2}(n-r))_{\mu} (\frac{1}{2}n)_{\delta} (t)^{-m} C_{\kappa}(D) g_{\kappa,\mu}^{\delta} C_{\delta}(I)/ \\ (\frac{1}{2}r)_{\kappa} C_{\kappa}(I) m! k!)$$

where $\Omega = \Phi^{-1} \beta A_{22} \beta'$ and

where $D = \frac{1}{2} \Phi^{-\frac{1}{2}} \beta A_{22} \beta' \Phi^{-\frac{1}{2}}$.

Case 2

$$(3.27) \quad M_2(t) = (\Gamma_q(\frac{1}{2}n) / \Gamma_q(\frac{1}{2}r)) (-1)^{-\frac{1}{2}(n-r)} q_1^{-\frac{1}{2}(n-r)} q_{I-P}^{\frac{1}{2}n} \\ \cdot \sum_m \sum_k \sum_{\mu} \sum_{\kappa} \Sigma_{\delta}((\frac{1}{2}(n-r))_{\mu} (\frac{1}{2}n)_{\delta} (\frac{1}{2}n)_{\kappa} \varepsilon_{\kappa, \mu}^{\delta} C_{\kappa}(P) C_{\delta}(I) (t)^{-m} / \\ (\frac{1}{2}r)_{\kappa} C_{\kappa}(I) m! k!)$$

where $P = \text{diag}(\rho_1^2, \dots, \rho_q^2)$ and $\mu = (m_1, \dots, m_q)$, $m_1 \geq m_2 \geq \dots \geq m_q \geq 0$, $\sum m_i = m$ is a partition of the integer m into not more than q parts, $\delta = (\delta_1, \dots, \delta_q)$, $\delta_1 \geq \delta_2 \geq \dots \geq \delta_q \geq 0$ and $\sum \delta_i = m + k$.

Proof

Let $E = A_{11} - A_{12} A_{22}^{-1} A_{21}$ and $B = A_{12} A_{22}^{-1} A_{21}$ then $Q = \text{tr} R^{-1} (I - R) = \text{tr} B^{-1} E$. Now for case 1 where $X^{(2)}$ is a set of fixed variables we have that E has a Wishart distribution $W(\Phi, n-r)$ and B has a noncentral Wishart distribution $W(\Phi, \Omega, r)$ (Anderson (1958)) with $\Omega = \Phi^{-1} \beta A_{22} \beta'$ ($q \leq r$)

Thus

$$(3.28) \quad M_1(t) = E(e^{tQ}) = E(e^{t \text{tr} B^{-1} E}) \\ = \int \int e^{t \text{tr} B^{-1} E} w(E; \Phi, n-r) g(B) dB$$

where $g(B) = w(B, \Phi, r) e^{-\frac{1}{2} \text{tr} \Omega} {}_0F_1(\frac{1}{2}r; \frac{1}{2} \Omega \Phi^{-1} B)$, $B > 0$ and $w(B, \Phi, r)$ is the density of the Wishart distribution (see Anderson (1958)).

In (3.28) let

$$M = (2\Phi)^{-\frac{1}{2}} E (2\Phi)^{-\frac{1}{2}}$$

$$N = (2\Phi)^{-\frac{1}{2}} B (2\Phi)^{-\frac{1}{2}}$$

with

$$J(E, B, M, N) = |2\Phi|^{(q+1)}, \text{ then}$$

$$\begin{aligned}
 3.29) \quad M_1(t) &= E(e^{t \text{tr} MN^{-1}}) \\
 &= (\Gamma_q(\frac{1}{2}r) \Gamma_q(\frac{1}{2}(n-r)))^{-1} \iint e^{t \text{tr} MN^{-1}} e^{-\text{tr} \frac{1}{2} \Omega |M|^{\frac{1}{2}(n-r-q-1)}} \\
 &\quad |N|^{\frac{1}{2}(r-q-1)} e^{-\text{tr} M - \text{tr} N} {}_0F_1(\frac{1}{2}r; DN) dM dN
 \end{aligned}$$

where $D = \frac{1}{2} \Phi^{-\frac{1}{2}} \beta A_{22} \beta' \Phi^{-\frac{1}{2}}$

Now integrating with respect to M yields

$$\begin{aligned}
 3.30 \quad \int e^{-\text{tr}(I - tN^{-1})M} |M|^{\frac{1}{2}(n-r-q-1)} dM \\
 &= \Gamma_q(\frac{1}{2}(n-r)) |I - tN^{-1}|^{-\frac{1}{2}(n-r)} \\
 &= \Gamma_q(\frac{1}{2}(n-r)) |N|^{\frac{1}{2}(n-r)} (-1)^{-\frac{1}{2}(n-r)} q_t^{-\frac{1}{2}(n-r)} q \quad |1 - t^{-1}N|^{-\frac{1}{2}(n-r)}
 \end{aligned}$$

Now

$$|1 - t^{-1}N|^{-\frac{1}{2}(n-r)} = \sum_{m=0}^{\infty} \sum_{\mu} \binom{\frac{1}{2}(n-r)}{\mu} {}_{\mu}C_{\mu}(t^{-1}N)/m!$$

where $\mu = (m_1, \dots, m_q)$, $m_1 \geq m_2 \geq \dots \geq m_q \geq 0$, $\sum m_i = m$.

Thus we have

$$\begin{aligned}
 3.31 \quad M_1(t) &= (\Gamma_q(\frac{1}{2}r))^{-1} \int (-1)^{-\frac{1}{2}(n-r)} q_t^{-\frac{1}{2}(n-r)} q |N|^{\frac{1}{2}(n-q-1)} e^{-\frac{1}{2} \text{tr} \Omega} \\
 &\quad e^{-\text{tr} N} \sum_m \sum_{\mu} \sum_k \sum_{\kappa} \binom{\frac{1}{2}(n-r)}{\mu} {}_{\mu}(t)^{-m} C_{\mu}(N) C_{\kappa}(DN) / (\frac{1}{2}!)_{\kappa} k! m! \, dN
 \end{aligned}$$

Now integrating over N and using Lemma 3.5 yields the result (3.26).

To find the moment generating function for case 2 notice that $M_1(t)$ depends on the set of variables $X^{(2)}$ only through A_{22} . Thus conditionally (for $X^{(2)}$ fixed) the moment generating function of Q is given by $M_1(t)$. To find the unconditional moment generating function $M_2(t)$, multiply $M_1(t)$ with the marginal density of A_{22} (which is $w(A_{22}, \Sigma_{22}, n)$) and then integrate over A_{22} using Lemma 3.4. This yields

$$\begin{aligned}
 3.32 \quad M_2(t) &= (\Gamma_q(\frac{1}{2}n) / \Gamma_q(\frac{1}{2}r)) (-1)^{-\frac{1}{2}(n-r)} q_t^{-\frac{1}{2}(n-r)} q \\
 &\quad \Sigma_{22}^{-1} \frac{1}{2} n | \Sigma_{22}^{-1} + \beta' \Phi \beta |^{-\frac{1}{2}n} \sum_m \sum_{\mu} \sum_{\kappa} \sum_{\delta} \binom{\frac{1}{2}(n-r)}{\mu} \binom{\frac{1}{2}n}{\delta} t^{-m} \binom{\frac{1}{2}n}{\kappa} g_{\kappa \mu}^{\delta} C_{\kappa} \\
 &\quad \cdot ((\beta' \Phi^{-1} \beta + \Sigma_{22}^{-1})^{-1} \beta \Phi \beta) C_{\delta}(I) / (\frac{1}{2}r)_{\kappa} C_{\kappa}(I) m! k! \,)
 \end{aligned}$$

where for case 2, $\Phi = \Sigma_{11.2}$.

Finally using the fact that if $\Sigma\Lambda = I$ then $\Sigma_{11.2} = \Lambda_{11}^{-1}$, $\Sigma_{12}\Sigma_{22}^{-1} = -\Lambda_{11}^{-1}\Lambda_{12}$ and $\Lambda_{22}^{-1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$ we obtain that $\beta\Sigma_{11.2}\beta + \Sigma_{22}^{-1} = \Lambda_{22}$ and $(|\Sigma_{22}||\Lambda_{22}|)^{-\frac{1}{2}n} = (|\Sigma_{11}||\Sigma_{11.2}|)^{\frac{1}{2}n} = |I-P|^{\frac{1}{2}n}$

with $P = \text{diag } \{\rho_1^2, \dots, \rho_q^2\}$ i.e. the squares of the population canonical correlation

coefficients. Similarly

$$C_K((\beta'\Phi^{-1}\beta + \Sigma_{22}^{-1})^{-1}\beta'\Phi^{-1}\beta) = C_K(P)$$

This yields the result (3.27).

The central moments of Q for cases 1 and 2 follows easily from (3.26) and (3.27), respectively.

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